Design Alternatives for the Evaluation of Design Science Research Artifacts

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ABSTRACT
Within a consideration of cost effectiveness the evaluation of design science research artifacts is of major importance. In the past, a plenitude of approaches has been developed for this purpose – partly artifact-specific, partly artifact-neutral. Nonetheless, there is a lack of a comprehensive overview over existing methods as well as a systemization of those with regard to fundamental structuring criteria. The paper at hand surveys existing methods and introduces a framework that equally supports the designer and the user of artifact evaluation approaches. Subsequent to the embedding of the framework into the design science research process two exemplary application scenarios are described.

Keywords
design science research, evaluation, artifact, morphological analysis, framework

1. INTRODUCTION
Even before evaluation took root as an established concept in last century's early 60ies it had been a crucial element of validating, revising, and controlling projects, products, and programs. Evaluation plays a vital role in both everyday life and science. WOTTAWA & THIERAU [34, pp. 22 ff.] portray the evolution of evaluation from its first rudimentary appearance in the primeval society to its broad dissemination in today's life. The selection of a certain stone that served best for working on some other material in the Stone Age can be seen as one of the first examples for evaluation. Here, already, objective criteria supported the assessment of a certain object concerning its suitability for a certain purpose. Due to its complexity and the variety of application areas today, it is difficult to clearly delineate the term evaluation [11, 28, 34]. A potential definition is offered by SCRIVEN [25, p. 85]:

"Evaluation is the process of determining the worth, merit, or significance of entities; and evaluations are the outcome of that process. Evaluation may be external or internal, or a mix of these; and it may be quantitative or qualitative, or a mix of these. It is strongly although not always sharply distinct from explanation."

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This definition may, however, further be endorsed with characteristics from a different definition provided by BOHLA [3, p. 299]. He defines evaluation as follows:

"Objective and systematic collection of information about a program, project, or instructional material for its improvement. (More recently in literature, evaluation is being defined as the systematic investigation of the worth or merit of an object; e.g., a program, project, or instructional material.)"

Admittedly, not only defining but particularly accomplishing an evaluation proofs to be difficult. To name but a few objectivity, comparability, and traceability of an evaluation represent major challenges [12, p. 36]. Thanks to the various interrelated factors that have to be considered when carrying out an evaluation the process is highly complex. In order to operate this complexity an adequate framework is required that supports a structured proceeding of the evaluation intentions.

The paper at hand addresses the subject of evaluation in design-oriented research and offers a framework that enables structuring and documenting the evaluation process alongside its fundamental characteristics. At the same time a comprehensive overview over existing design-oriented evaluation methods is given.

The remainder of this paper is structured as follows. The second chapter provides a theoretical background briefly introducing design science research (DSR) and existing approaches for the evaluation of artifacts within design-oriented information systems (IS) research. The main body of the paper (chapter three) is devoted to the introduction of a framework containing various alternatives for the evaluation of design science research artifacts. Guidelines on how to use the framework as well as exemplary configurations are given (chapter four). Finally, the paper concludes with a short summary and identifies areas for further research (chapter five).

2. THEORETICAL BACKGROUND
2.1 Design science research
IS research is largely characterized by two paradigms, namely behavioral science and design science. While behavioral science concentrates on the development and verification of theories, design science focuses on the development of solutions for practical problems and, thereby, on accomplishing utility [14, p. 76, 17, p. 256]. According to HEVNER ET AL. and MARCH & SMITH, the outcomes of a construction process under the design science paradigm are constructs, models, methods, and instantiations [14, p. 79, 17, p. 257]. The development process of DSR artifacts

commonly cover the phases 'problem identification', 'requirements specification', 'design', 'evaluation' and 'communication' [13, p. 62, 14, pp. 82-90, 17, pp. 254 ff.]. In order to actually achieve utility when developing a DSR artifact two fundamental requirements must be fulfilled: relevance and rigor. Relevance is achieved when the artifact actually addresses a real business need. Rigor, in contrast, is assured by appropriately applying the existing body of knowledge, i.e. theoretical foundations and methodologies. Against this background in particular the evaluation of artifacts represents a key challenge. In the following four major contributions are briefly introduced.

2.2 Existing approaches for artifact evaluation

Not least because of economic considerations a high importance is ascribed to the evaluation of DSR artifacts. Hence, a variety of different evaluation approaches has been developed, partly originating from the IS discipline itself, partly inherited from its parent disciplines business administration (and, thereby, likewise sociology) and computer science. Subsequently, four frameworks that categorize and systemize existing evaluation approaches in a different manner are introduced.

NIEHAVES & PFEIFFER [22]

The authors investigate the applicability of structuralism for evaluating conceptual models and, thereby, mainly focus on the artifact type model. They structure their contribution based on three research questions: "What are the elements of a holistic approach to evaluating [information technology] IT artifacts (Q1)?", "Is a holistic approach to determining the quality of a conceptual model available yet (Q2)?", and "How can structuralism contribute to a holistic approach to evaluating conceptual models (Q3)?" [22]. The proposed matrix-framework spans two dimensions: 'artifact' and 'evaluation'. The artifact dimension covers the abovementioned artifact types construct, model, method, and instantiation (cf. section 2.1), whereas the evaluation dimension includes the artifact structure, evaluation criteria, and evaluation approaches. By means of a literature search selected research results are identified and thereon assigned to the fields of the framework. The main body of the contribution is dedicated to the introduction of an approach for evaluating information models using structuralism.

FETTKE & LOOS [10]

FETTKE & LOOS develop a multiperspective framework for the evaluation of reference models. They, too, focus on the artifact type model and motivate the development of their framework with the fact that modeling represents a time-consuming and quite error-prone endeavor. Despite the hence evident significance of a high reference model quality for the deduction of individual models, FETTKE & LOOS constitute a lack of adequate approaches for the evaluation of reference models in the IS and related disciplines. The proposed framework covers 15 different perspectives that correspond to different evaluation methods [10, p. 83]. This framework, too, spans two dimensions, namely 'foundation of quality criteria' and 'research method'. Concerning the dimension foundation of quality the authors further distinguish between theory-driven and ad hoc. As to the dimension research method they differentiate between analytical and empirical. FETTKE & LOOS briefly explain all of the 15 perspectives and name respective advantages and disadvantages. The authors do not claim completeness for their framework, but rather call for further researchers to take this framework as a basis for additional work on evaluation methods.

FRANK [11]

The approach proposed by FRANK – like the previous one – explicitly focuses on the evaluation of conceptual models, and, in particular, on the evaluation of reference models. FRANK as well deplores the absence of appropriate evaluation methods in the IS discipline and suggests a comprehensive reference structure based on four fundamental perspectives. He considers an economic, a deployment, an engineering, and an epistemological perspective crucial for a holistic assessment of reference model quality [11, p. 123] (a closer consideration of these perspectives follows in section 3.1). The author complements the different perspectives with a plenitude of aspects and criteria as well as relations amongst these [11, pp. 125-135]. This turns FRANK'S approach into very practicable and sophisticated one.

SIAU & ROSSI [26]

SIAU & ROSSI conduct an extensive comparative literature analysis on the subject of evaluation approaches for modeling methods and, thereby, too, consider only one of the DSR artifacts. Like PFEIFFER & NIEHAVES as well as FETTKE & LOOS, however, the authors cite a plethora of different methods. Beyond, they criticize the uncontrolled growth of new methods developed by eager practitioners and researchers [26, p. 2]. Many of these methods – the authors constitute – are lacking a theoretical foundation and an empirical validation of their utility. SIAU & ROSSI, therefore, conduct a systematic survey of existing approaches, describe their functionality, and name strengths and weaknesses. Subsequently, they develop a framework for systemizing the collected approaches. The framework obtains the two dimensions ontology and epistemology and, thereby, facilitates a classification of the methods in terms of the underlying philosophy of science [26, p. 13].

Each of the four frameworks provides an interesting and illuminating contribution to the topic of evaluation. Nonetheless, these approaches either focus on gathering existing evaluation methods and providing an overview or are restricted to the evaluation of merely one DSR artifact type. The paper at hand, thus, firstly aims at providing a general framework for DSR artifact evaluation and at delineating in which phase of the design process evaluation plays a role. Furthermore, it strives to support IS researchers in conducting evaluation intentions by offering various design alternatives. The following section introduces the framework, its components, and functionality.

3. DESIGN OPTIONS

3.1 Structuring of design variables and values

As motivated in section 1 and outlined based on selected examples in section 2.2 a variety of different evaluation methods and criteria as well as frameworks for their systemization already exist. The paper at hand interprets the evaluation process of a DSR artifact like the artifact itself and, thus, as a design object (cf. sections 2.1 and 3.2). Those variables that are subsequently introduced as relevant for the evaluation of DSR artifacts can partly be assigned to the IS discipline (and herein in particular to DSR) and partly to the disciplines of business administration and sociology, which exhibit a long tradition of evaluation. Variables as well as their respective values are consolidated and depicted in a morphological field [24, p. 794] (cf. Figure 1).

Variable	Value				
Approach	Qualitative		Quantitative		
Artifact Focus	Technical	Organizational	Strategic		
Artifact Type	Construct	Model	Method	Instantiation	Theory
Epistemology	Positivism		Interpretivism		
Function	Knowledge function	Control function	Development function	Legitimization function	
Method	Action research	Case study	Field experiment	Formal proofs	
	Controlled experiment	Prototype	Survey		
Object	Artifact		Artifact construction		
Ontology	Realism		Nominalism		
Perspective	Economic	Deployment	Engineering	Epistemological	
Position	Externally		Internally		
Reference Point	Artifact against research gap	Artifact against real world	Research gap against real world		
Time	Ex ante		Ex post		

Figure 1. Variables and values for the evaluation of DSR artifacts.

In the following, variables and values for the evaluation of DSR artifacts that are presented in figure 1 are explained in detail.

Approach

The evaluation approach represents the way in which characteristics of the evaluation object are assessed (e.g., facts, claims, and assumptions).

- *Qualitative*. Characteristics of the evaluation object are not appraised on a numerical, but on a value basis, and "emphasizes the description and understanding of the situation behind the factors" [8, p. 205]. However, qualitative data can be converted into numerical data through codification [3, p. 304].
- *Quantitative*. In the case of a quantitative evaluation characteristics of the evaluation object are, in contrast, assessed on a numerical basis [3, p. 304]. Note, however, that it is of course possible to evaluate both ways – quantitatively and qualitatively – and that a mixed method approach may provide even deeper insights into the quality of the evaluation object.

Artifact Focus

DSR artifacts can support a company's transformation process on the technical, organizational, and strategic level [19]. This variable addresses the artifact's intended context of use, not the development context.

- *Technical*. Technical DSR artifacts are for example routing algorithms, communication protocols, image processing algorithms, hardware designs, or robots.
- *Organizational*. Organizational DSR artifacts are for example process models and grammars, methods for organizational redesign, or accountability matrices.
- *Strategic*. DSR artifacts on the strategic level are for example designs for decision support systems, roadmap development methods, or balanced scorecards.

Artifact Type

A number of classifications for DSR artifacts has been suggested in the past [e.g., 4, 32]. This contribution follows the widely recognized classification by MARCH & SMITH [17, pp. 256 ff.] and HEVNER ET AL. [14, p. 76] and distinguishes between instantiation, construct, method, and model. Lately, there have been various debates about whether or not theories should be interpreted as DSR artifacts as well [13, 15, 30]. The authors consider theories to be a fundamental part of DSR – not only as theoretical basis, but also as possible outcome of a design process. Thus, theories are added to the list of DSR artifacts.

- *Construct*. Constructs offer syntax and semantic for a domain to describe the problem and/or solution (e.g., classification systems, language standards, or ontologies).
- *Model*. By means of models relations between construct elements can be explicated and certain aspects of reality are abstracted (e.g., through meta models or reference models).
- *Method*. Methods represent algorithms and proceedings for the solution of specific problems or classes of problems (e.g., approaches for business process modeling or software development).
- *Instantiation*. An instantiation represents a concrete realization of a construct, model, or method. It demonstrates feasibility and enables researchers to actually test their concepts under real world conditions and learn more about the real world.
- *Theory*. GREGOR distinguishes between five different types of theory in the area of IS research [13, pp. 619 ff.]. In general, theories describe cause and effect relations. For an in-depth discussion of the role of theory in DSR please refer to VENABLE [30] and KUECHLER & VAISHNAVI [15].

Epistemology

Epistemology represents a research subarea of the philosophy of science and is concerned with how 'true knowledge' is achieved and how special characteristics of the researching scientist affect the process of knowledge acquisition [2, p. 201]. Positivism and interpretivism represent two epistemological extreme positions.

- *Positivism*. A positivist epistemological attitude implies that the evaluation of an artifact leads to the same objective results, regardless of the individual characteristics of the evaluating person [26, p. 12].
- *Interpretivismus*. Under an interpretive stance, in contrast, it is assumed that the results of an artifact evaluation highly depend on the individual characteristic of the evaluating subject [26, p. 12].

Function

According to STOCKMANN [28, pp. 66 f.], evaluation serves different general functions that are, however, closely interweaved with each other. Subsequent general functions can be distinguished.

- *Knowledge function*. Evaluation should be carried out for the acquisition of knowledge that may put management decisions

on a rational basis. The aim is to derive knowledge relevant for steering decisions from the assessment of predefined assessment criteria [24, pp. 67 f.].

- *Controlling function*. The knowledge about structures and processes won during the evaluation process can be used for control. First of all, it can be examined whether the criteria defined in the planning phase have been fulfilled. Criteria may, for example, be efficiency, effectiveness, acceptance or lastingness. Even if the control function is not the first and foremost aim of evaluation, it is always indirectly or directly part of the process [24, p. 68].
- *Development function*. The learning processes associated with the development function are based on the insights won by the control and the knowledge function. These insights facilitate the discourse between involved stakeholders and allows for enhancing the considered artifact. The learning function plays an important role when conducting a formative evaluation [24, p. 68].
- *Legitimization function*. The last function of evaluation is the legitimization of an artifact. The knowledge achieved during the evaluation enables a traceable documentation of inputs and outcomes of the artifact construction process. However, there is the possibility that artifacts are ex-post legitimized with suitable criteria [24, pp. 68 f.].

Method

For the evaluation of IS artifacts different methods and technologies have been suggested (a comprehensive and in-depth discussion of each method is outside the scope of this work). The herein introduced evaluation methods for DSR artifacts can be found in SIAU & ROSSI [26] and MARCH & STOREY [18].

- *Action research*. Action research represents a qualitative research approach [1, p. 94]. Research object and researcher are not separated, but much more influence each other mutually in their development. Another characteristic of action research is the cooperation of researchers and the users of the artifact (e.g., employee of a company in which a model is applied and evaluated) [26, p. 11].
- *Case study*. Case studies depict a broadly used means for describing or analyzing real world phenomena. Sources for information gathering are e.g. interviews, documents, or measurements. A decisive characteristic of case studies is the deliberate admitting of uncontrolled influences that emerge from the context (e.g. a company) an artifact is evaluated in [3, p. 297, 26, pp. 10 f.]. For further information and detailed guidelines for evaluation with case studies please refer to YIN [35].
- *Field experiment*. A field experiment is also carried out in real world surroundings (e.g., in a company). The experimentalist manipulates and controls some attributes of the surroundings (independent variables). As many characteristics as possible are measured and observed in order to gather the influence of the manipulated attribute on the results (dependent variable) (i.e. the effect of independent on the dependent variables) [26, p. 11].
- *Formal proofs*. When applying formal proofs, rules are defined that have to be considered during the construction process. PARSONS & WAND [20] describe an example of the production, evaluation and application of such rules. Formal proofs constitute an adequate method for DSR artifacts evaluation [18, p. 728].
- *Controlled experiment*. In a controlled experiment different situations are compared that differ in only as to one characteristic (experiment variable, independent). The effect of this characteristic on the dependent variables of the experiment is observed. The aim is to isolate a certain effect as far as possible. All remaining characteristics that may probably have an effect on the dependent variables must be held as steady as possible. The comparison of a control group with an experiment group is typical [26, p. 9].
- *Prototype*. An evaluation based on a prototype facilitates the assessment of a solution's suitability for a certain problem by implementing the solution generically [18, p. 726]. An example of an evaluation with a prototype is described in LEE ET AL. [16]. Evaluation with a prototype represents an adequate evaluation method for DSR artifacts [18, p. 728].
- *Survey*. By conducting a survey, information is collected through interviewing representatives of a certain target group. Questions can relate to subjective and objective issues, whereas the answers are always subjectively and only limitedly verifiably [26, pp. 8 f.].

Object

CROSS [9, pp. 52 f.] and WINTER [33, p. 471] both distinguish between design research on the one hand and design science on the other. The object of evaluation, therefore, is either an artifact or its construction process.

- *Artifact*. A DSR artifact itself can be object of evaluation [14, pp. 85 ff., 17, p. 261, 33, pp. 471 f.].
- *Artifact construction*. The process in which DSR artifacts are constructed may as well be evaluated [21, pp. 70 ff., 33, pp. 471 f.].

Ontology

Ontology represents a further part of the philosophy of science discipline that deals with the essence of being and the basic structure of reality [26, p. 12]. It is analyzed 'what' exists and 'how' things exist. Basically, two major attitudes can be distinguished: realism and nominalism.

- *Realism*. Realists hold the opinion that the world exists regardless of human perception, i.e. regardless of whether the world is thought or spoken about [2, p. 202]. Accordingly, for realists reality (objects as well as structures) is objectively given.
- *Nominalism*. Nominalists, in contrast, postulate that the world is not given in the form of objective facts, but rather construed in inter-subjective communication and other social processes [6, p. 5].

Perspective

FRANK [11] describes four perspectives out of which reference models can be evaluated. All perspectives contain aspects which do not exclusively apply to reference models and are discussed in similar form also at other places (cf. section 2.2). Therefore, the suggested perspectives can also be used for the evaluation of other artifact types.

- *Economic perspective*. The economic perspective considers costs, use and expenditures which originate from the construction on the one and the use of DSR artifacts on the other hand [11, pp. 124-130].
- *Deployment perspective*. The deployment perspective considers comprehensibility and acceptance aspects of implementing and using DSR artifacts [11, pp. 130 f.].
- *Engineering perspective*. The engineering perspective addresses the manner (e.g., used modeling or programming

languages, degree of description of components, or integrity and flexibility aspects) in which the DSR artifact is constructed [11, pp. 131-133].

- *Epistemological perspective*. The epistemological perspective considers the basis of the construction process as to the underlying philosophy of science (e.g., methodical foundation, generality, or objectivity of evaluation) [11, pp. 133-136].

Position

A DSR artifact can be evaluated by people who are involved in the construction, or by exterior people.

- *Externally*. External evaluation is carried out by people, who are not involved in the construction of the artifact to be evaluated [3, p. 300].
- *Internally*. An internal evaluation is carried out by the designers of the artifact [3, p. 301].

Reference point

According to BUCHER ET AL. [5, p. 75], three different reference points can be taken when evaluating results of IS research.

- *Artifact against research gap*. The artifact is evaluated in respect of correct design either against established or against previously defined requirements. An employment under real world conditions is not realized. The correctness of the derived research gap remains unreflected.
- *Artifact against real world*. In this case the suitability of the construed artifact is assessed through employment in the real world. Thereby, it becomes obvious if the artifact actually endows utility as a solution for the original problem. The adequacy of the research gap is at the same time implicitly reflected.
- *Research gap against real world*. This approach is of a subordinated meaning in the context of the IS research and is primarily addressed under the behavioral paradigm. Statistical quality criteria may exemplarily be mentioned as measures for the representativeness of research results.

Time

PRIES-HEJE ET AL. [23] distinguishes the time, when the evaluation of an artifact is conducted.

- *Ex ante*. With ex ante evaluation, candidate systems or technologies are evaluated before they are chosen and acquired or implemented.
- *Ex post*. With ex post evaluation, a chosen system or technology is evaluated after it is acquired or implemented.

In order to ensure a consistent and coherent quality assessment of DSR artifacts, evaluation needs to be well planed and integrated into the DSR process. The subsequent section discusses how this aim can be achieved.

3.2 Configuring evaluation design criteria in the DSR process

Starting from the very early contributions by SIMON [27] as well as MARCH and SMITH [17] on design as a science a variety of IS researchers has provided valuable approaches for DSR strategies, practices, and processes. One of the most famous and most cited articles is the one written by HEVNER ET AL. called 'Design Science in Information Systems Research' [14]. However, not only HEVNER and his colleges but also TAKEDA ET AL. [29] WALLS ET AL. [31], CARLSSON [7], and a number of further IS researchers, not explicitly named herein, offer valuable advice on how to proceed when developing a DSR artifact. One of the newest and a very comprehensive article is provided by PEFFERS AT AL. [21]. In their contribution the authors accumulate a large proportion of existing work on DSR. We thus decide to take their framework as the basis for the explanation on how to configure evaluation design criteria in the DSR process. PEFFERS ET AL. [21, P. 54] describe a DSR methodology (DSRM, cf. Figure 2) for the construction and presentation of DSR artifacts in the IS context [21, p. 48]. Following a detailed discussion and consolidation of several approaches by PEFFERS ET AL. [21, pp. 55 ff.] within this paper the DSRM is referred to as a reference process for the DSR artifact construction. It is the aim to align the morphological field for DSR evaluation (cf. Section 3.1) to that process.

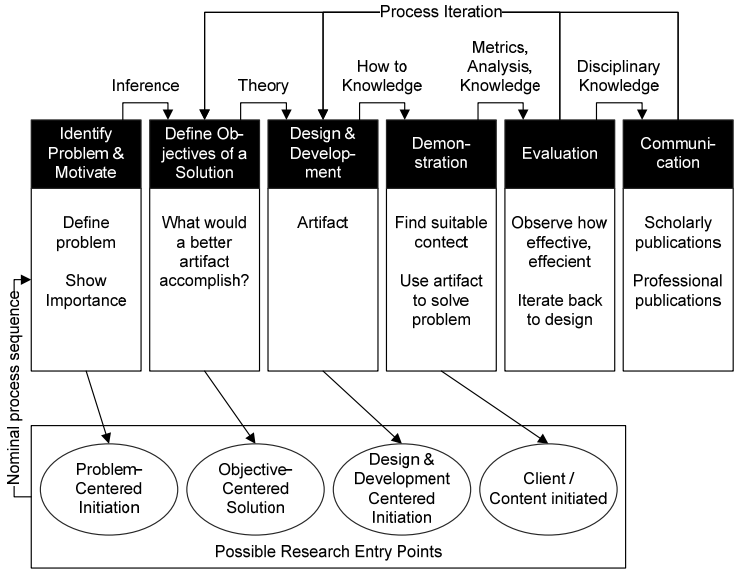

Figure 2. Design science research method [21].

DSRM explicitly addresses the evaluation of DSR artifacts in activity 5 [21, p. 56]. This activity compares observations made during the use of the specific artifact to design objectives defined in activity 2. The following enumeration briefly describes the use of the morphological field (i.e. the configuration of particular variables) during the respective DSRM activities [21, pp. 52-56]. However, the chosen links between configured variables and DSRM activities are not meant exclusively but exemplarily. It is aspired to exemplify that the design and execution of an evaluation project (i.e. the configuration and use of the morphological field) may be present along the whole DSRM process.

- *Activity 1: Identify problem and motivate*. Both process and result of identifying a research problem is driven and influenced by the researcher's philosophical mindset. In this activity, the variables ontology and epistemology may be configured. Regarding an identified research problem, decisions concerning the artifact type and focus, object, and reference point may further be made.
- *Activity 2: Define objectives of a solution*. Due to influencing evaluation metrics and methods (i.e. interviews, experiments), the definition of design objectives demands for configuring the variables approach, function and perspective.
- *Activity 3: Design and development*. At the beginning of activity 3, the variable time needs to be configured, as later on, an ex ante evaluation will no longer be possible. A special case is given, if an ex ante evaluation is conducted along multiple DSRM iterations. But even then, a decision concerning the variable time should be made before the execution of activity 3 in the first iteration.
- *Activity 4: Demonstration*. This activity comprises the use of the designed artifact in at least one case. If the variable method is configured with case study, for example, this activity may be part of the evaluation. In this case, activity 5 would comprise only analysis and documentation of the case study.
- *Activity 5: Evaluation*. For this activity, a completely configured morphological field must be present. Especially the configurations of time, method, and object influence the evaluation's execution.
- *Activity 6: Communication*. The communication of a DSR artifact as a research result in a scientific context comprises both a detailed description of the artifact itself and its evaluation (i.e. progress, analysis and results). Hence, the evaluation's design (i.e. the configuration of the morphological field) influences the communication activity as well.

4. Uses cases for the morphological field

4.1 Decision support

The previous section presents the DSRM approach [21] and the equivalent alignment of the morphological field (cf. Section 3.1). The field variables are used to structure and document the evaluation design. A further benefit of the presented approach may be the outcome of an analysis of DSR artifacts along the given evaluation dimensions (i.e. variables of the morphological field). For that purpose, the morphological field could be used as an analysis scheme for classifying DSR artifacts. Given an adequate base of analyzed artifacts (i.e. analysis of scientific work that presents DSR artifacts and the artifact), one could use these data for decision support during an artifact's evaluation design. The following example is supposed to illustrate this kind of use: A researcher with a nominalistic (ontology) and positivistic (epistemology) attitude wants to evaluate a method (artifact type) for documenting a knowledge progress (function). An analysis of significant correlations between particular field configurations may support the decision for an ex ante (time) evaluation by means of formal proofs (method).

A further result of analyzing DSR artifacts along the given evaluation dimensions may be the evaluation of these dimensions themselves. That evaluation would aim at the amount and diversity of dimensions: The overall objective is to get a set of dimensions that is as small as possible and as large as necessary. A significant correlation of particular configuration combinations might be a hint for redundant dimensions.

4.2 Evaluating DSR artifacts

In a special issue of the IS journal "Management Information Systems Quarterly", MARCH & STOREY [18, p. 728] present five DSR artifacts included in the issue as well. Figure 3 and Figure 4 show field configurations for the artifacts presented BY LEE ET AL. [16] and PARSONS & WAND [20].

Variable	Value				
Approach	Qualitative		Quantitative		
Artifact Focus	Technical	Organizational	Strategic		
Artifact Type	Construct	Model	Method	Instantiation	Theory
Epistemology	Positivism		Interpretivism		
Function	Knowledge function	Control function	Development function	Legitimization function	
Method	Action research	Case study	Field experiment	Formal proofs	
	Controlled experiment	Prototype	Survey		
Object	Artifact		Artifact construction		
Ontology	Realism		Nominalism		
Perspective	Economic	Deployment	Engineering	Epistemological	
Position	Externally		Internally		
Reference Point	Artifact against research gap	Artifact against real world	Research gap against real world		
Time	Ex ante		Ex post		

Figure 3. Exemplary configuration for LEE ET AL. [16].

Variable	Value				
Approach	Qualitative		Quantitative		
Artifact Focus	Technical	Organizational	Strategic		
Artifact Type	Construct	Model	Method	Instantiation	Theory
Epistemology	Positivism		Interpretivism		
Function	Knowledge function	Control function	Development function	Legitimization function	
Method	Action research	Case study	Field experiment	Formal proofs	
	Controlled experiment	Prototype	Survey		
Object	Artifact		Artifact construction		
Ontology	Realism		Nominalism		
Perspective	Economic	Deployment	Engineering	Epistemological	
Position	Externally		Internally		
Reference Point	Artifact against research gap	Artifact against real world	Research gap against real world		
Time	Ex ante		Ex post		

Figure 4. Exemplary configuration for PARSONS & WAND [20].

LEE ET AL. [16] present a process grammar and a related method to generate and manage business process design alternatives. Furthermore, a supportive software prototype is presented. The evaluation is conducted by comparing several design alternatives generated by the prototype. Figure 3 shows the respective configuration of the herein proposed evaluation framework.

PARSONS & WAND [20] present a model for classification and rules that can be used for the identification of classifications. The evaluation is conducted by formal methods and expert interviews. Figure 4 again shows the respective configuration of the framework.

By using the morphological field in terms of a classification framework, both artifact reviewers and designers can compare DSR artifacts based on the variables and their configuration provided by the field. In that use case, the presented approach does not provide any ranking (e.g. “formal proofs are better than a field experiment). The field is only intended to provide some nominal scales to compare.

5. CONCLUSION AND FURTHER RESEARCH

The paper at hand outlines impact and importance of evaluation in design science research. We present a brief overview of evaluation in general and align our findings with the design science research methodology presented by PEFFERS ET AL. [21]. We emphasize that evaluation should not be seen as an isolated process activity but needs to be considered from the beginning of a design process.

Due to multiple aspects of evaluation in general, we argue the need for a structured evaluation design approach and present a morphological field that can be configured and used during the whole design process. For illustrating how to use that field, we describe a possible adaption in the DSRM process and exemplary configurations for particular DSR artifacts.

Some further research is already described in Section 4. A database that is built by classifying DSR artifacts along the presented evaluation dimensions might be used for both decision support during the design process and evaluation of the evaluation dimensions themselves. Given an appropriate amount of data, one could build a configuration tool for supporting the researcher (e.g. selecting appropriate evaluation method for a given configuration of the remaining variables).

Furthermore, we outline that the presented set of evaluation dimensions is not meant as a complete list but as a starting point for both discussion and extension. According to some basic principles (i.e. as small as possible, as large as necessary) the set of dimensions might not only extended or reduced by statistical analysis (cf. Section 4), but also by further literature reviews and practical needs.

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